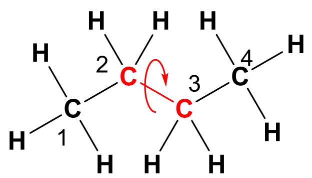
Construction of Conformational Energy Diagram of Butane

【Introduction】

The three-dimensional shapes of molecules result from many factors such as steric hindrance. A molecule may assume different shapes, called conformations, that are in equilibrium at room temperature (the conformational isomers are called conformers, emphasis on the first syllable). The systematic study of the shapes of molecules and properties from these shapes is stereochemistry. The field of stereochemistry is one of the central parts of organic chemistry and includes many important topics.

Butane ($CH_3CH_2CH_2CH_3$) is well-known to have four conformational isomers which are frequently referred to as a good example to study the correlation between the shape of a conformational isomer and its stability. In this lecture, you will draw computationally an energy diagram illustrating the change in energy that occurs with rotation about the C2–C3 bond in order to study the above-mentioned correlation a little.



⇒ Four conformational isomers will be found!

【Keywords】

・*isomer* (異性体) :

any one of a group of chemical substances which all have the same molecular formulae but that are structurally different in some way

・*conformational isomers* (配座異性体) :

isomers having different spatial arrangements of atoms of a molecule but have same bond connectivity and the same configuration

・*steric hindrance* (立体障害) :

hindrance of chemical action ascribed to the arrangement of atoms in a molecule

・*torsion angle* (ねじれ角) :

angle between the plane containing atoms ABC and the plane containing BCD in a nonlinear chain of atoms A-B-C-D

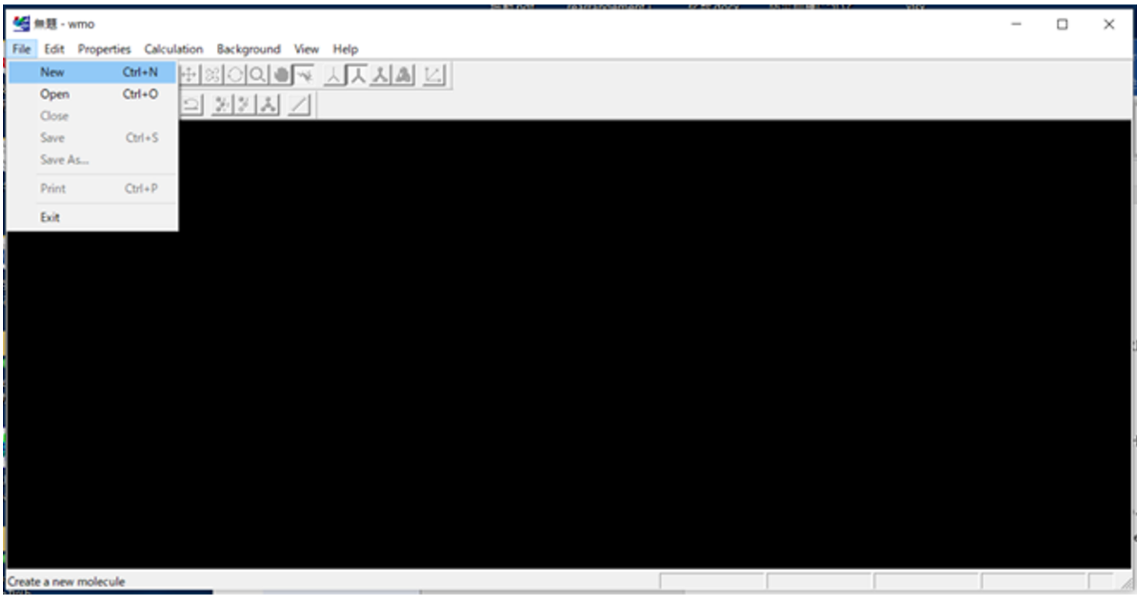
・*semi-empirical molecular orbital calculation* (半経験的分子軌道計算) *

* Now, you need not know the meaning of this word because it would be too difficult for students who have not studied quantum chemistry.

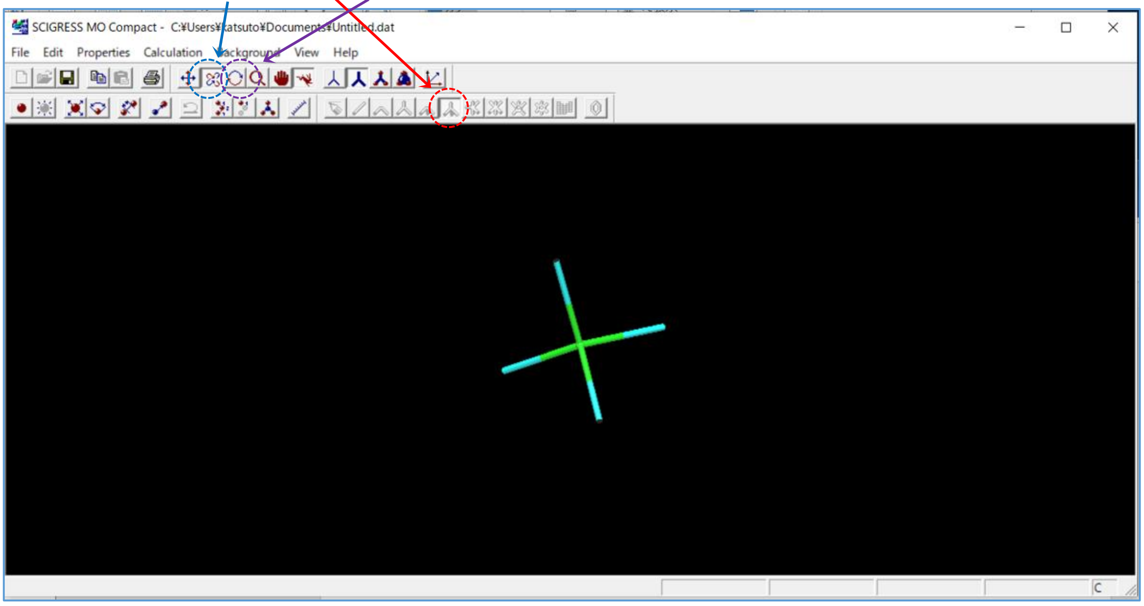
【Procedure to draw the energy diagram using a semi-empirical molecular orbital calculation program】

1. Double-click on the desktop icon labelled “SCIGRESS MO Compact 1.0 Standard”.

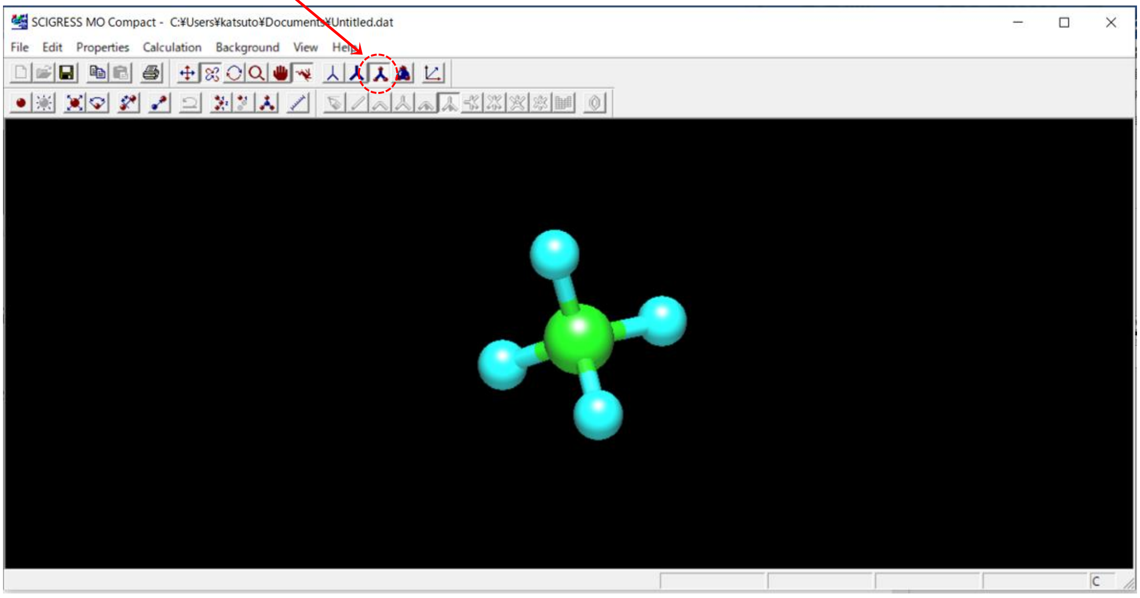
2. Select “New” from the File menu.



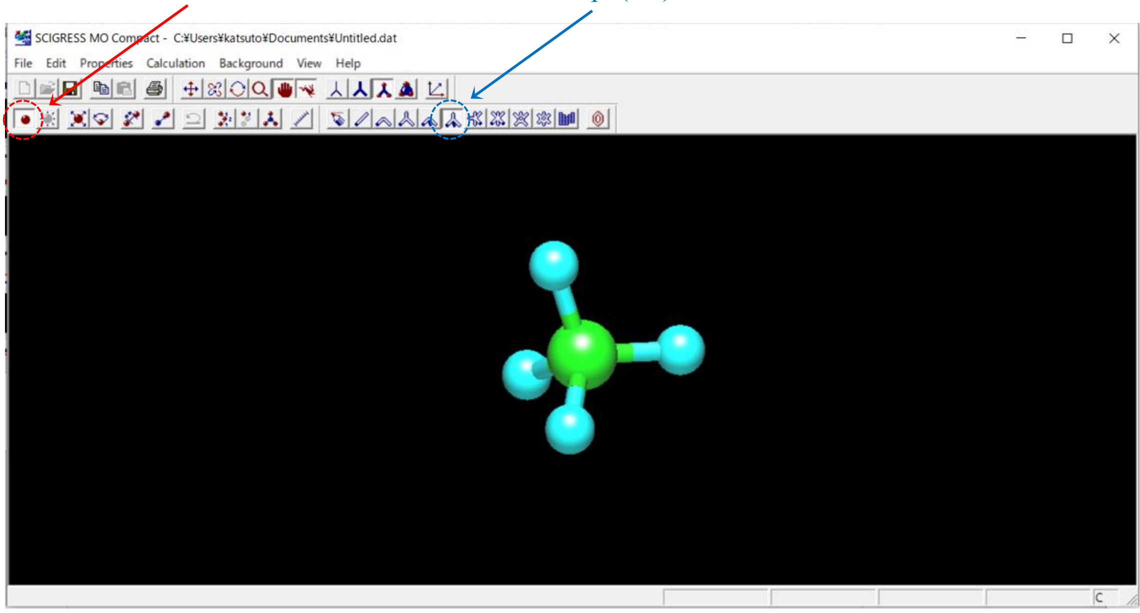
3. Click on the “sp3(3H)” button and then click on workspace. A 3D-model of “methane” appears on the workspace. You can rotate the model by moving the cursor on the workspace while clicking, after you click on the “rotation” or “rotation(z)” button.



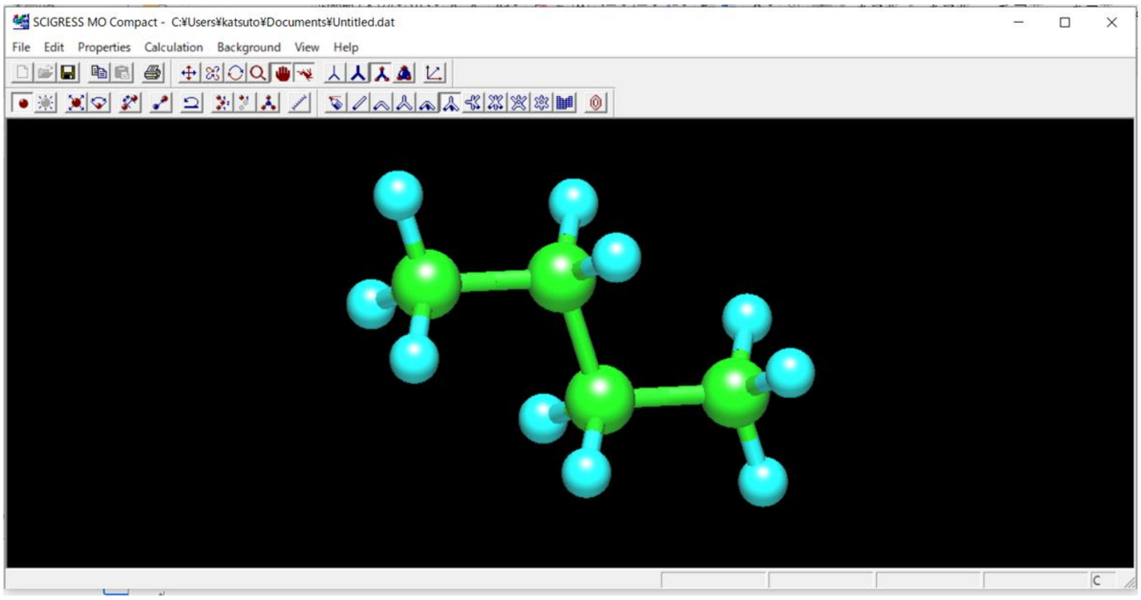
4. Click on the “ball & stick” button to represent atoms and covalent bonds as spheres and lines, respectively.



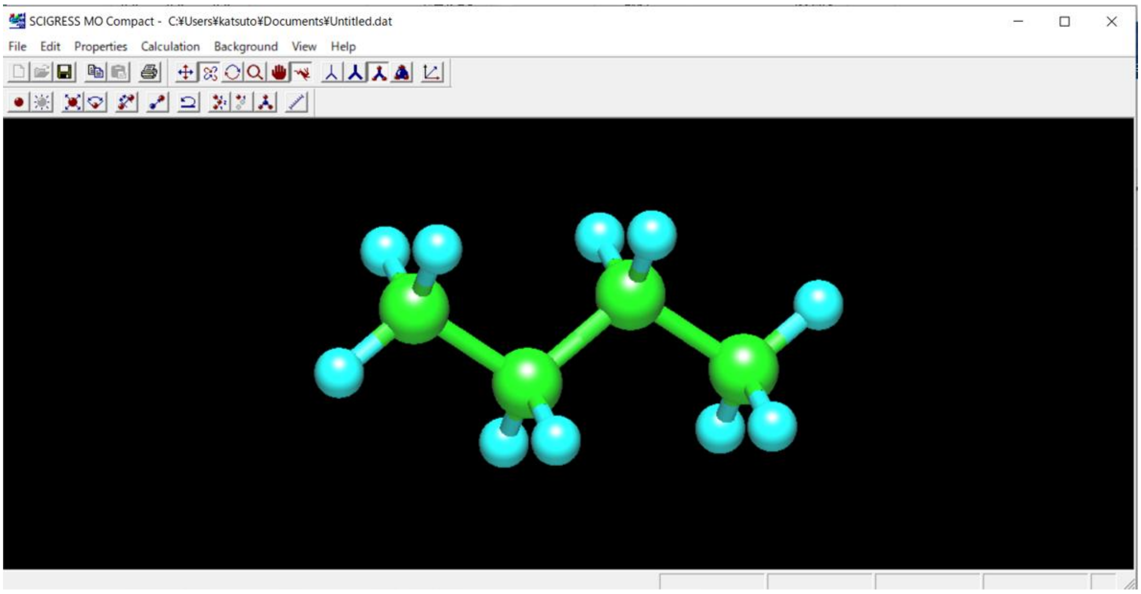
5. Click on the “add” button and then click on the “sp3(3H)” button.



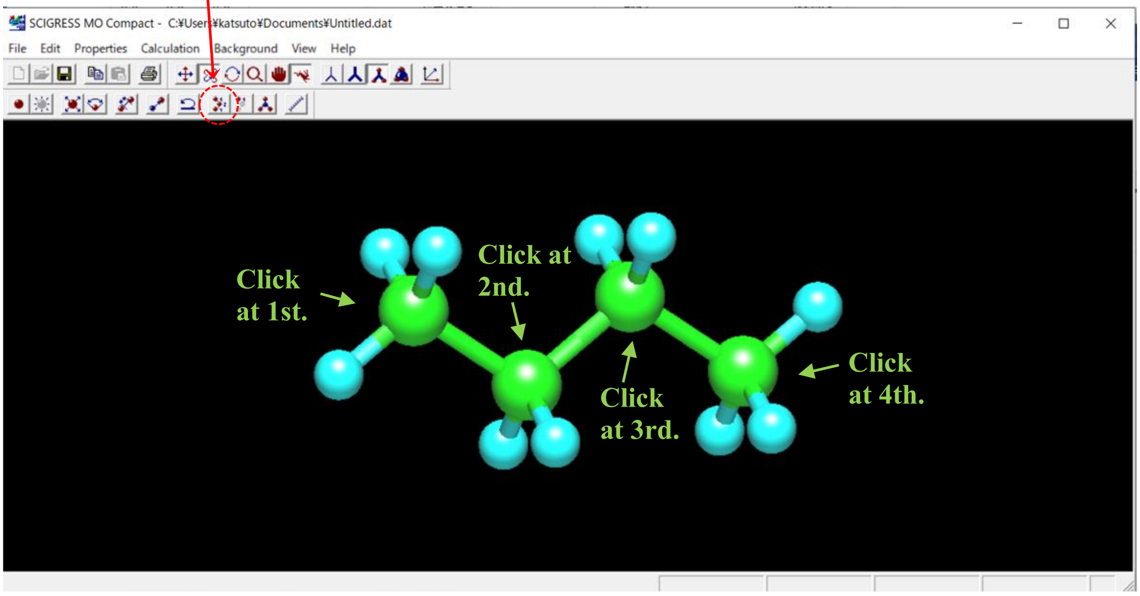
6. Repeat clicking on a light-blue sphere at the end of the 3D-model three times to make a "butane" structure in which light-green spheres and light-blue spheres represent carbon atoms and hydrogen atoms, respectively.



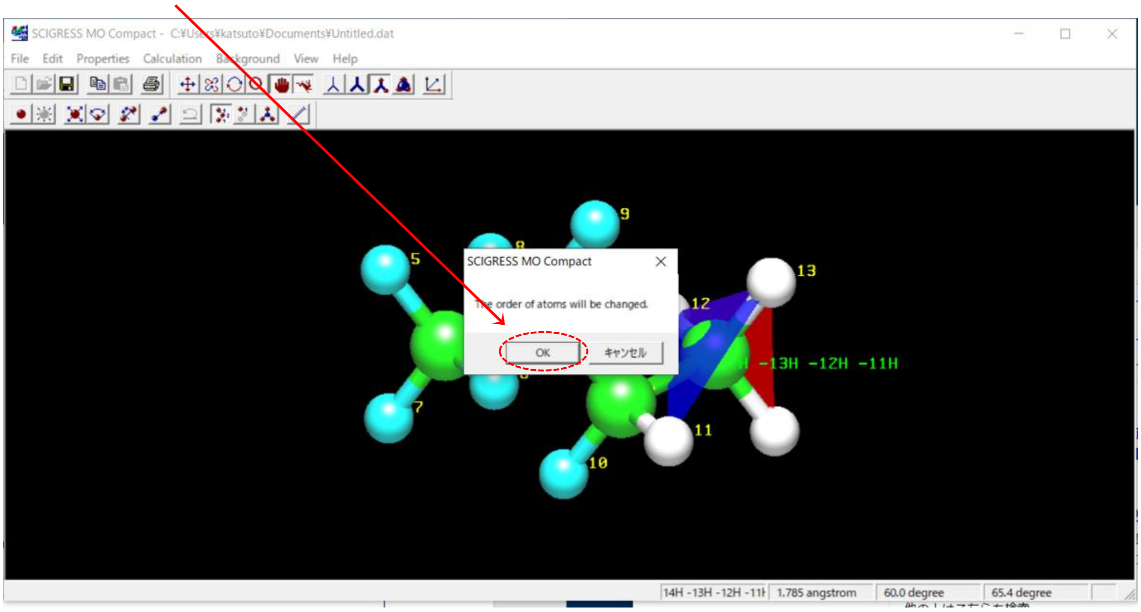
7. Rotate the 3D-model as shown in the figure below.



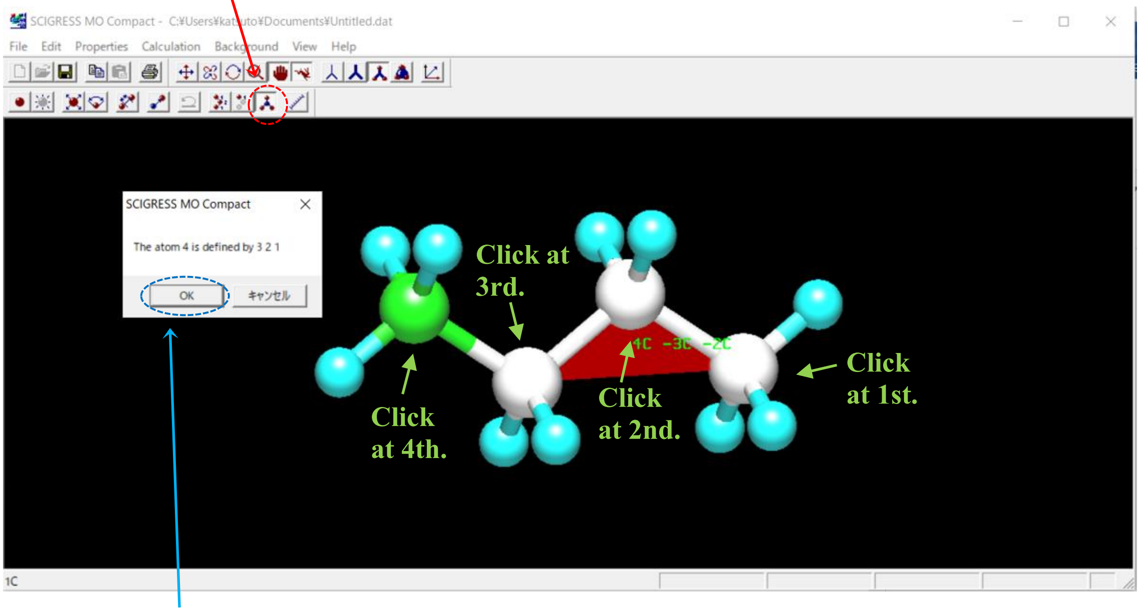
8. Click on the “renumber” button and then click on the four spheres representing carbon atoms in order from the left and ten spheres representing hydrogen atoms in any order subsequently.



9. Click on “**OK**” button.



10. Click on the "change znum" button and then click on the four spheres representing carbon atoms in order from the right.

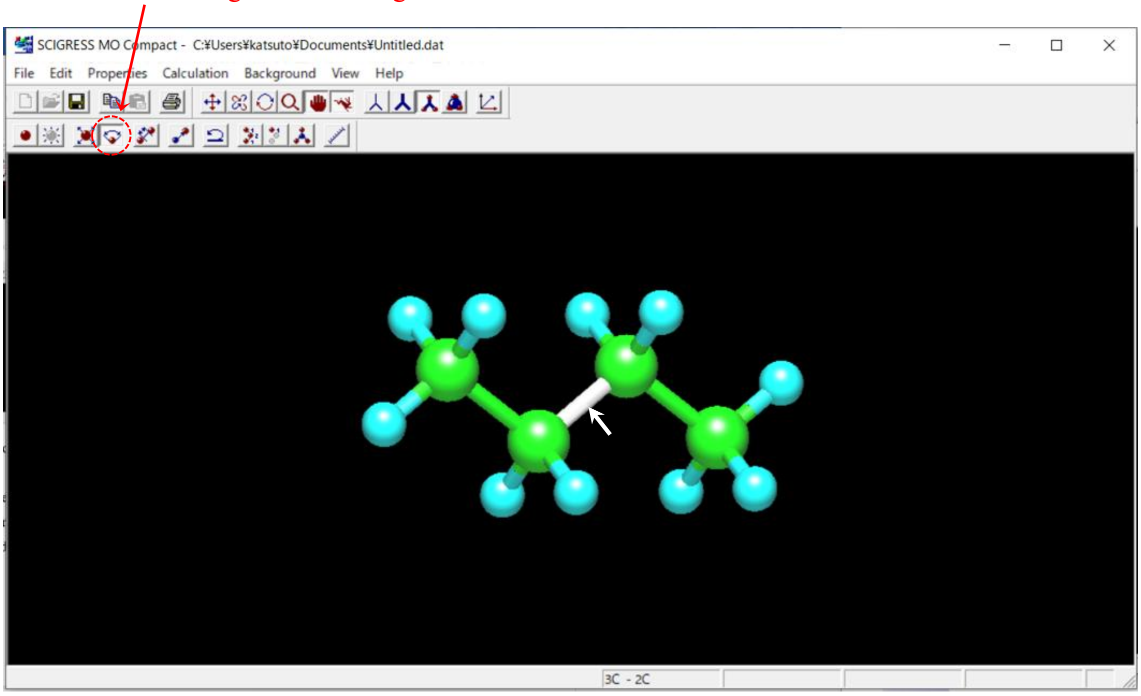


11. Click on "**OK**" button.

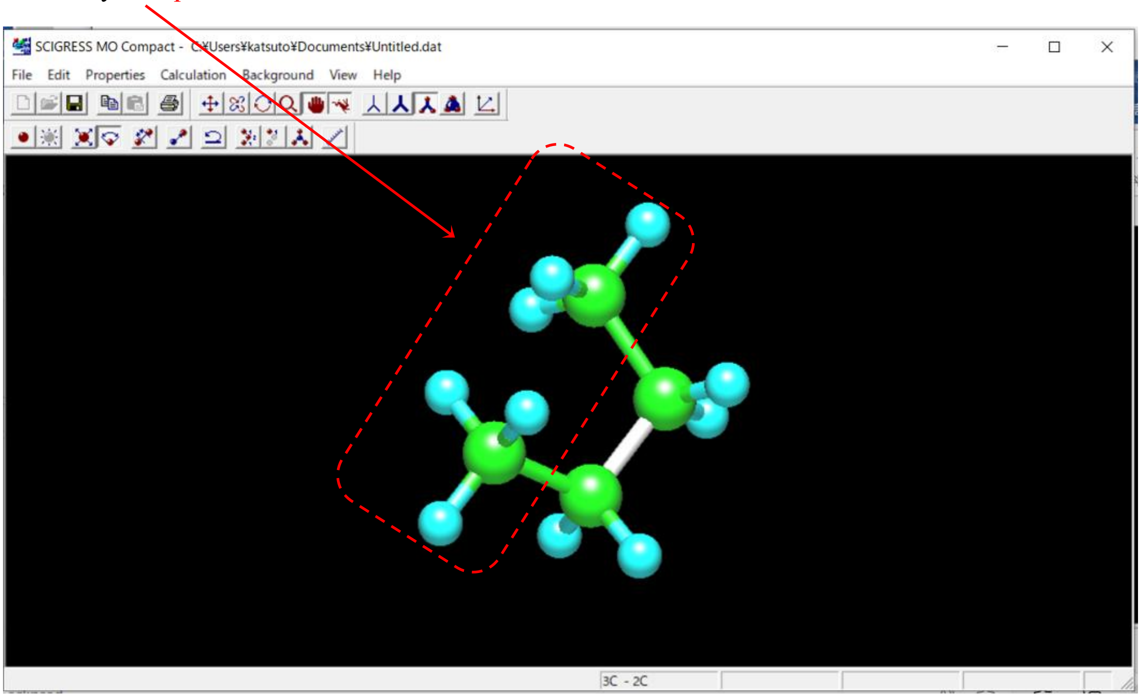
12. Create a folder named "**bond rotation in butane**" on the desktop.

13. Select "Save As..." from the File menu and then save the file with the name "**bond rotation in butane**" in the folder you created.

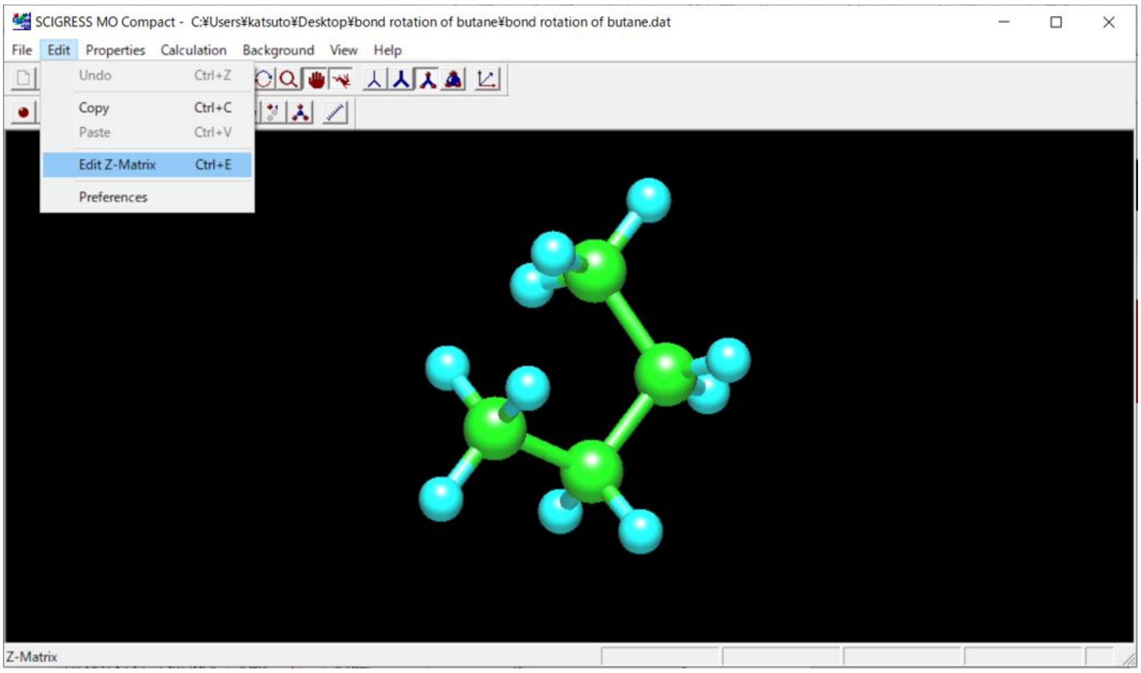
14. Select the "change torsional angle" button and then move the cursor on C2-C3 bond.



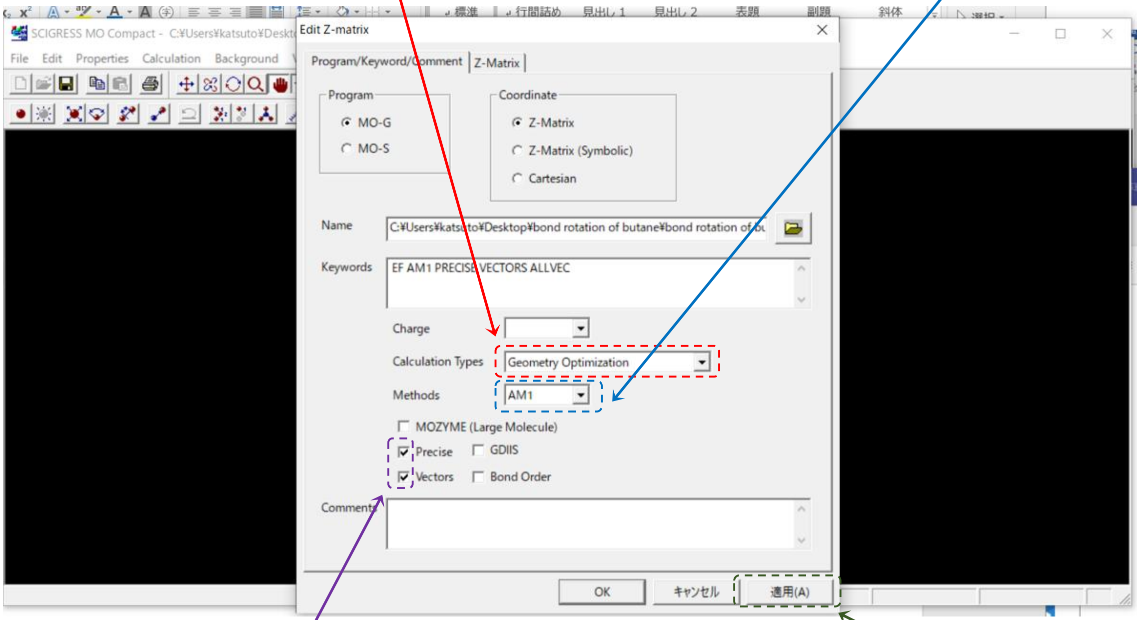
15. Move the mouse keeping its left button clicked to rotate the C2-C3 bond till two methyl groups are nearly “eclipsed”.



16. Select “Edit Z-matrix” from the Edit menu.



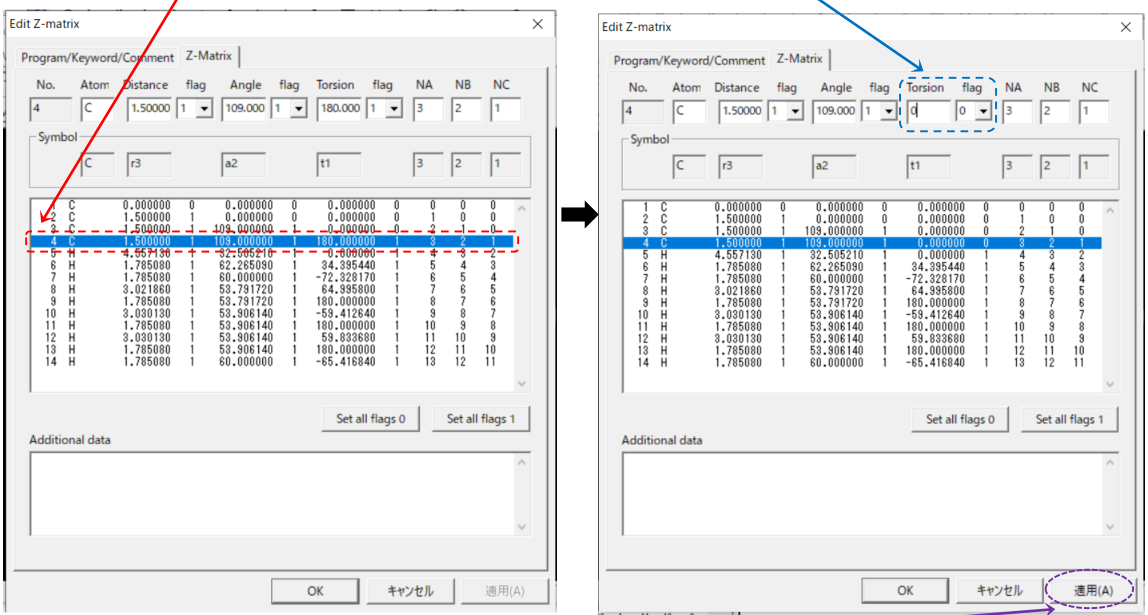
17. Select "Geometry Optimization" from the pull-down menu for Calculation Types and "AM1" from the pull-down menu for Methods, respectively.



18. Check the boxes for Precise and Vectors, respectively and then click on the "適用(A)" button.

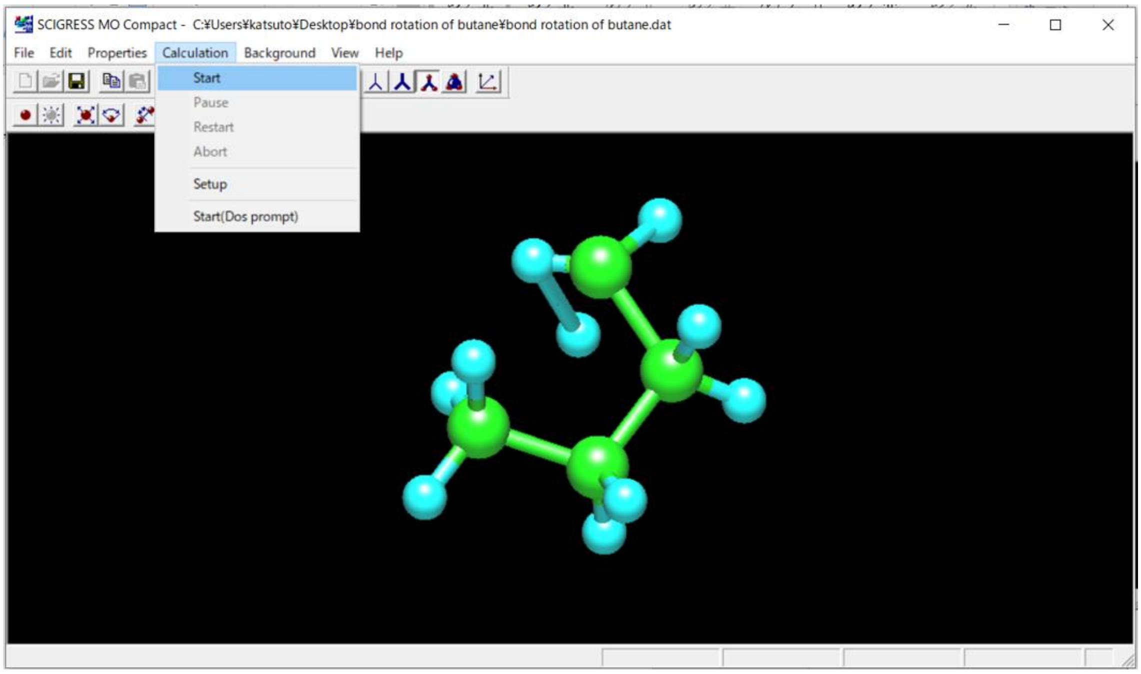
19. Select the "Z-matrix" tab.

20. Select the fourth row of the table and then change both the value of torsion angle and its flag number to 0.

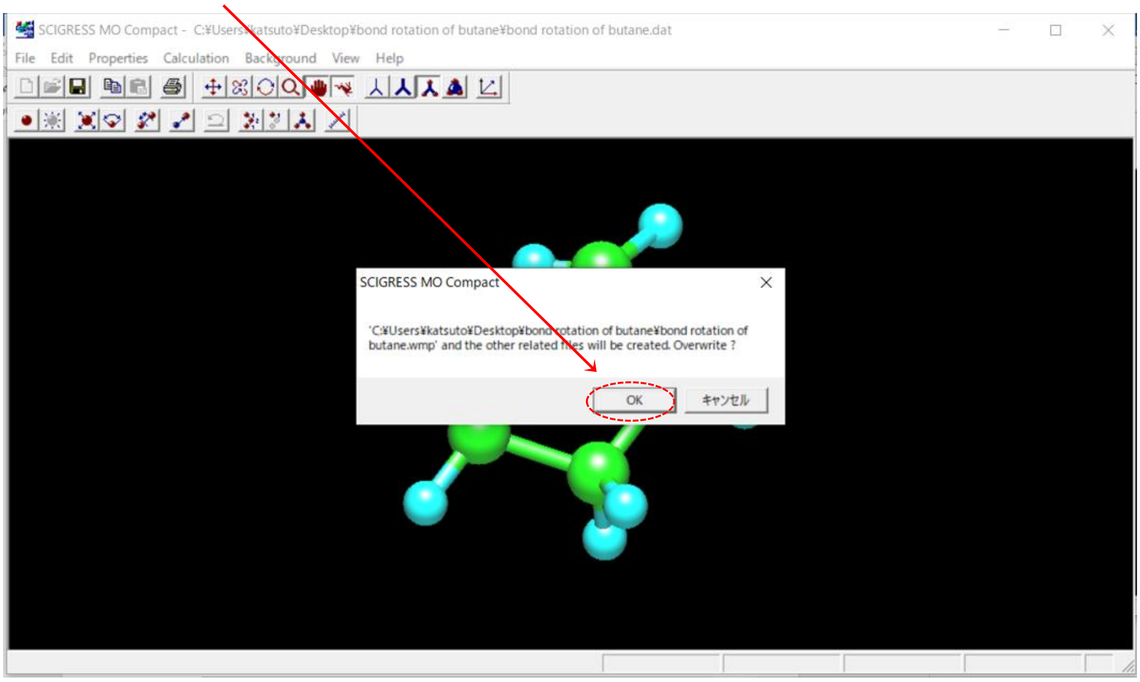


21. Click on the "適用(A)" button and then click on the "OK" button.

22. Select “Start” from the Calculation menu.

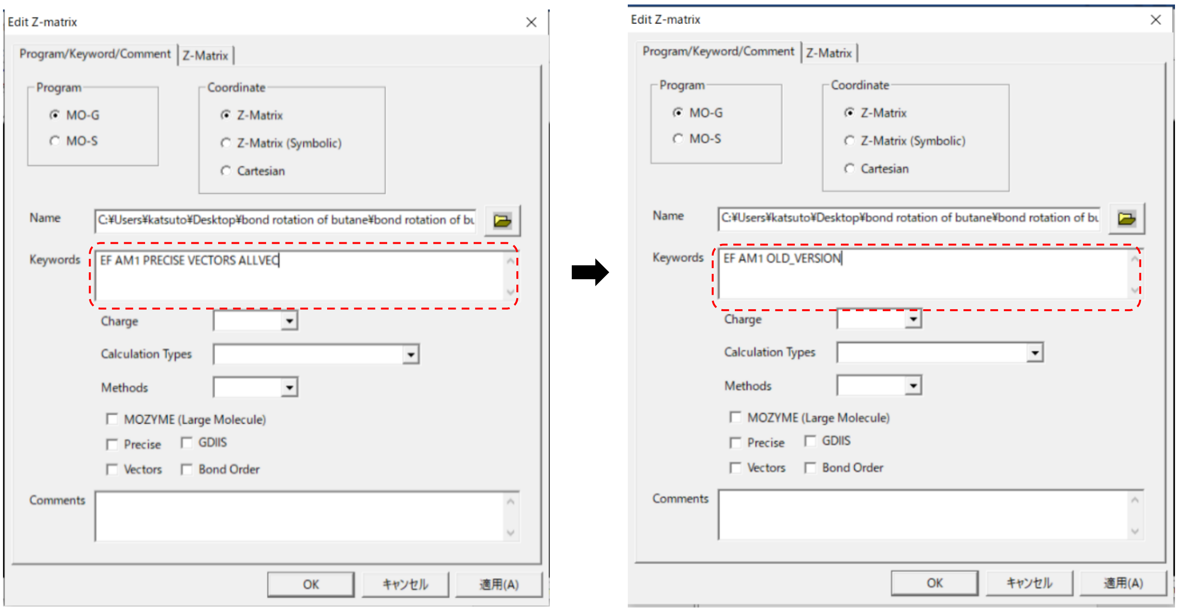


23. Click on the “OK” button and wait till a message “MO-G done” appears at the lower left.



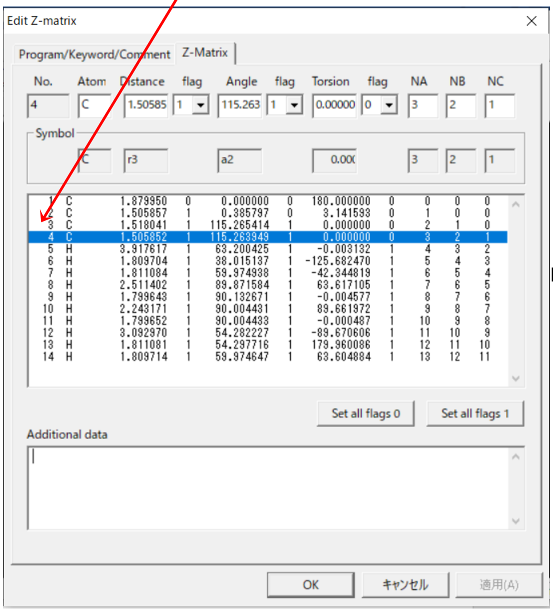
24. Select “Edit Z-matrix” from the Edit menu.

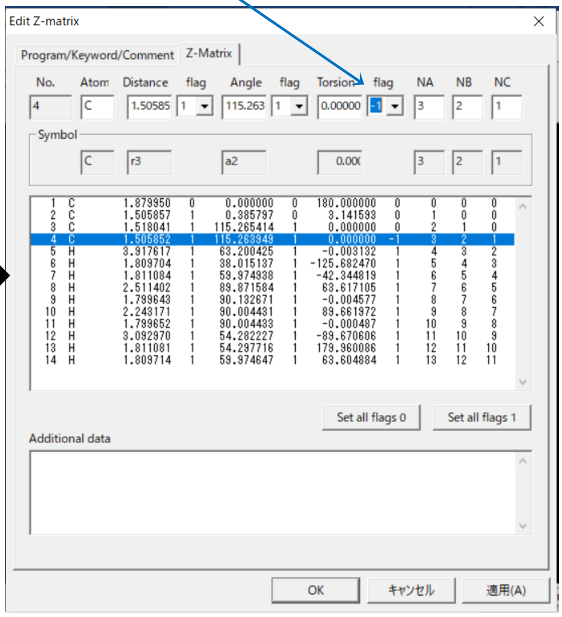
25. Replace keywords “PRECISE VECTORS ALLVEC” with “OLD_VERSION” and then click on the “適用(A)” button.



26. Select the “Z-matrix” tab.

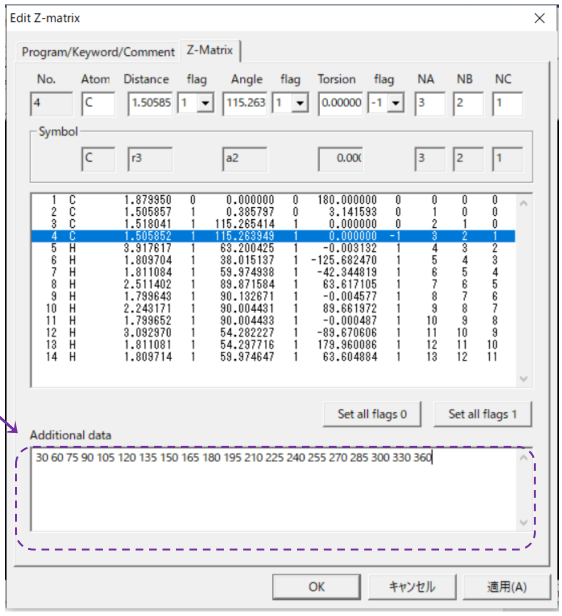
27. Select the fourth row of the table and then change the value of flag for the torsion angle to -1.



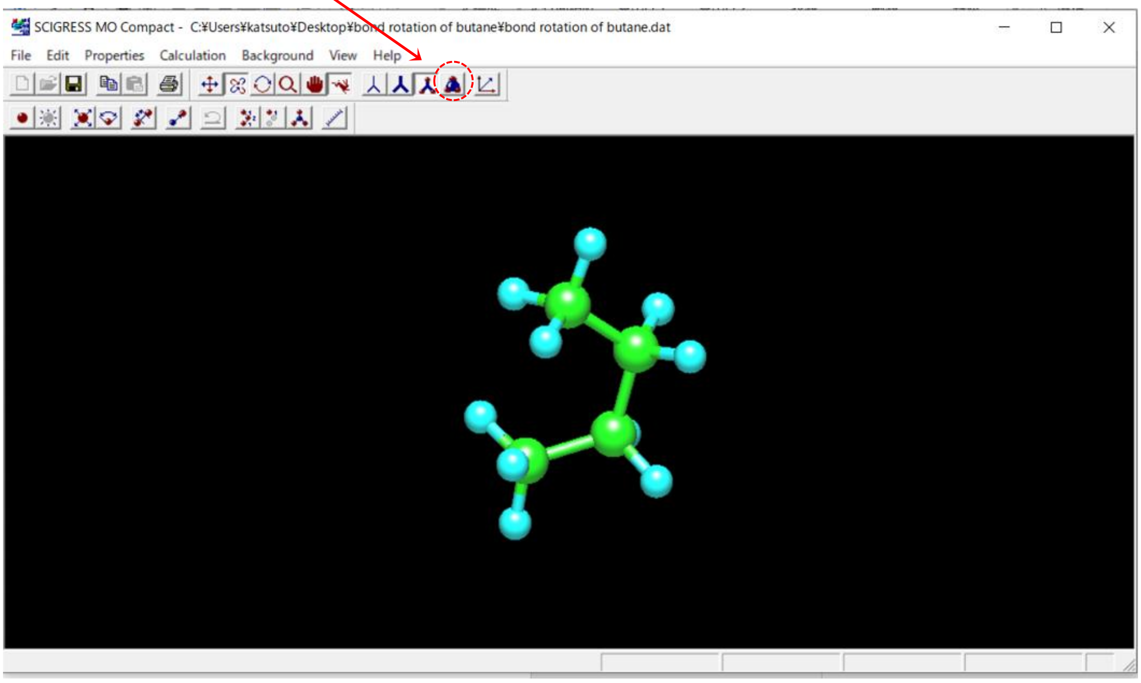


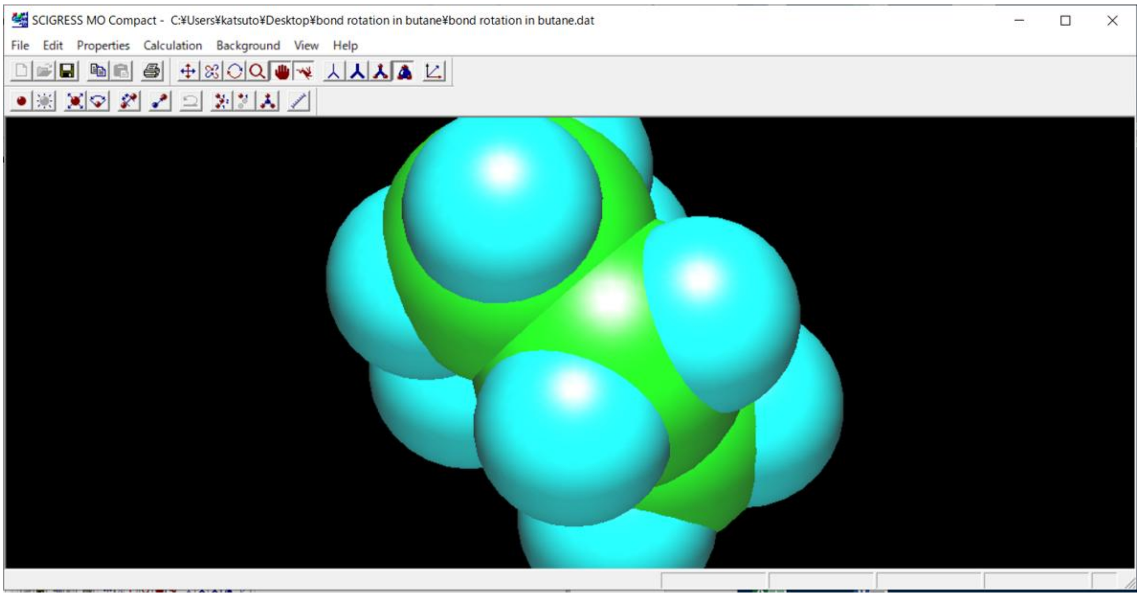
28. Input “30 60 75 90 105 120 135 150 165 180 195 210 225 240 255 270 285 300 330 360” in the additional data box.

29. Click on the “適用(A)” button and then click on the “OK” button.

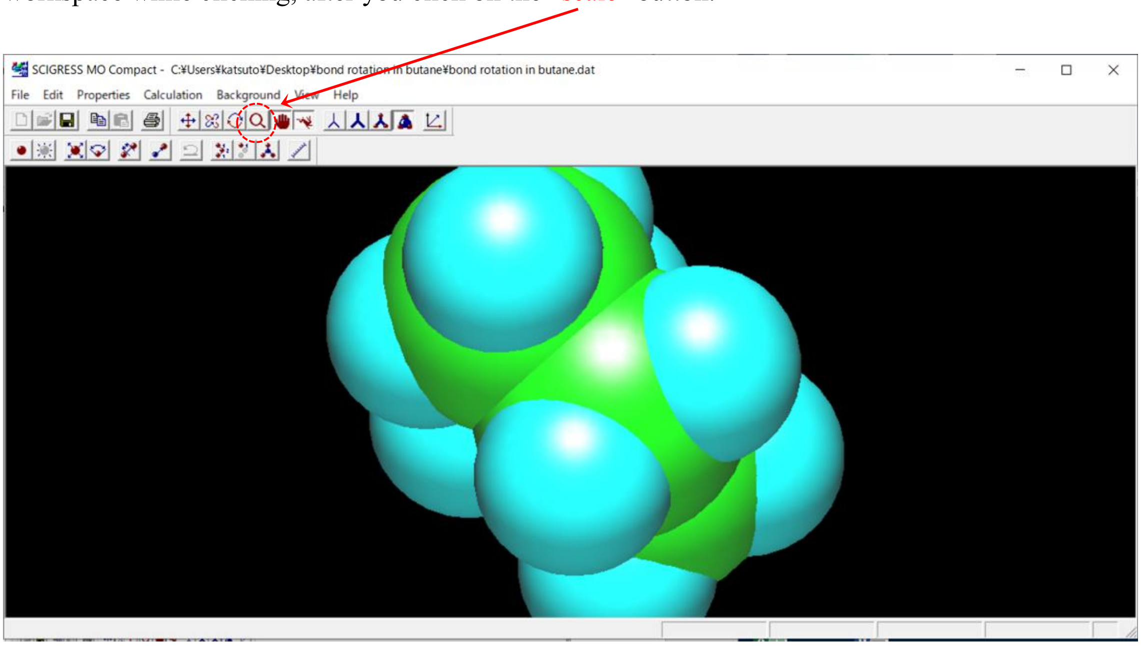


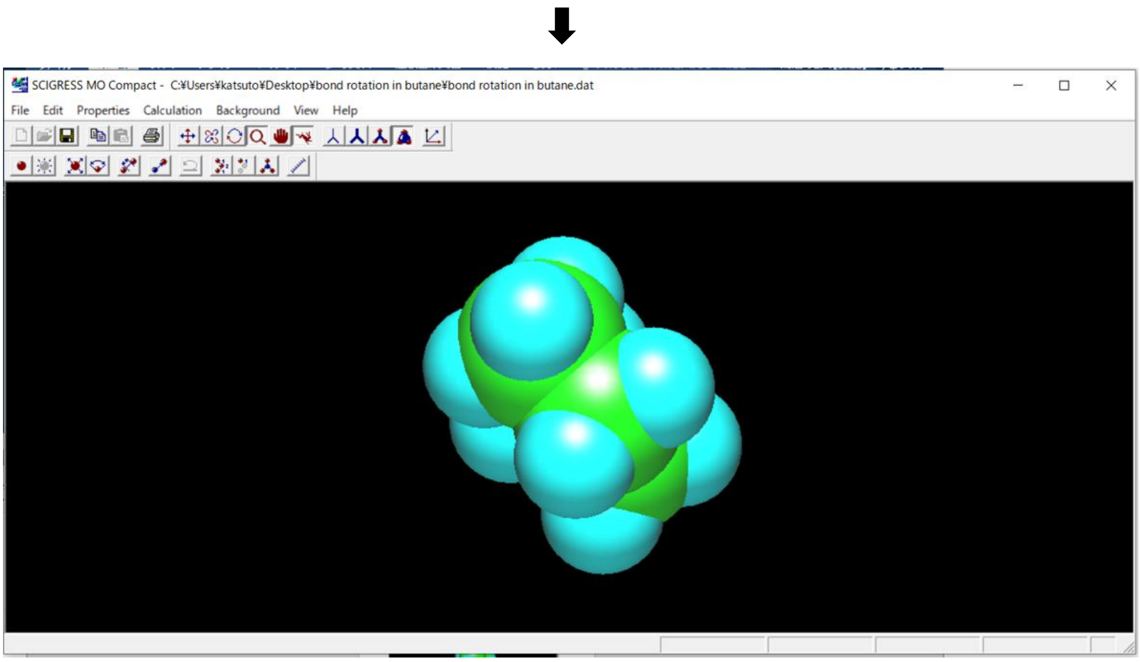
30. Click on the "spacefill" button to display the 3D-model where the atoms are represented by spheres whose radii are proportional to the radii of the atoms.





31. If the image is too large to fit on the workspace, you can reduce it by moving the cursor on workspace while clicking, after you click on the "scale" button.





32. Select "Start" from the Calculation menu.

33. Click on the "OK" button and wait till a message "MO-G done" appears at the lower left.

34.Select “Reaction” from the Properties menu. You can find an energy diagram on which the calculated heat of formation (HoF) is plotted versus the torsion angle made by C1-C2-C3-C4 in butane. The diagram illustrates the change in energy that occurs with rotation about the C2–C3 bond.

The diagram is not shown in this document. It is desirable that you try to click on various parts in the diagram to study the change in energy of butane associated with its conformational change. For example, ・ You can view the animation that conformation of butane changes accompanied with the bond rotation between C2 and C3 bond. ・ You can rotate 3D-models of butane with each HoF value in order to view it from various angles so that you consider congestion between atoms that increases (or decreases) in steric hindrance.

【Questions】

・Draw a Newman projection*, looking down the C2-C3 bond, of the most stable conformation of butane.

* You should look up “Newman projection” if you do not know the meaning of this word.

・Draw a Newman projection, looking down the C2-C3 bond, of the second stable conformation of butane.

・Draw a Newman projection, looking down the C2-C3 bond, of the most unstable conformation of butane.

・Answer the characteristics of the most stable conformation of butane, focusing on the steric hindrance.

・Answer the characteristics of the most unstable conformation of butane, focusing on the steric hindrance.

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